

PRACTICE MATERIAL • VERIFIED QUESTION BANK

Q.1. •ù8•ù8"%@ "%@‘É@•ù2 ‘ù?’ù0-1 "% 'i0"Ù- ("%" 'iM'ù> '©M" #“é2” “Y “Y?’Ù>‘ÉM'ù\$“ââ *”Ù0“iM”, 8•) ”Ù/“â 3 “ •ù •™.”Ù.” &“Y

A) 'OM' '\$u8'U%e* , *U0'eG'd & U5'eo'a 8IM'u>©?B IM'Y?'I@o'i')OBS#é' BM&(E % @ Y0'2 'ÉOR' d'U5'eo'a'R' U'P'F+VUS BQ Wf'E'o@a'mm

C) 'ÉM'u>e2?'l@òù •) #n'\$' / 'OM' '\$ù')•'m1 'YG.Y(' 0")* -Y- "7m' E%'%-% Jwvz?IB'B'6R' 0' F-Ü' Á B6AMWf'c'g' g'y'20'S' k'p'/%-E'U'Vd'OMC+

A) É'M'ù'è'è'I@'dù') 'y#''ù\$' / '©M' \$'ù.''''' 'YG·Y('0')*·Y·' 7''B''''Z''(ES'M'Y%12'1B3%'@·Y·F·2·O50'65'W5·ad'35'U·Q·è·a·u·'·Q·'·u·v·e·Q·W·e·b·

[illegible]

A) $\text{e}^{i(\frac{2\pi}{\hbar} \frac{m}{2} \frac{v^2}{2})} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(i k x) \exp(-i k v) \exp(-i k^2 \frac{m}{2} \frac{v^2}{2}) dk$

[illegible][illegible][illegible][illegible]

parameter (Case : Ó#sB•

- Q.19.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Cyclic Quadrilaterals & Tangent-Chord Angles, Problem #72): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits (Root B=792)
C) 'IM"Y?'%>'B 'Y5"â ") ("%."é""é('IM"Y">" > '©M" >'©M'B ("ù6"UB)S6P, %9."2. A=500) / Define value derived via quadratic calculus resolution boundary equality (Root D=1224)
- Q.20.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Triangle Centres: Incentre, Circumcentre, Orthocentre & Centroid, Problem #91): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) '©M" \$"ù,•)' "%@'é>'9 'YK '©B" > 'Y0"%G "Y>")> "%?" M'iG'i>B) / IM"Y?%>'B"Y5"â ") ("%."é""é('IM"Y">" > '©M" >'©M'B ("ù6"UB)S6P, %9."2. A=500) / Define value derived via quadratic calculus resolution (Root A=637)
C) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits boundary equality (Root D=1547)
- Q.21.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Triangle Centres: Incentre, Circumcentre, Orthocentre & Centroid, Problem #111): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) '©M" \$"ù,•)' "%@'é>'9 'YK '©B" > 'Y0"%G "Y>")> "%?" M'iG'i>B) / IM"Y?%>'B"Y5"â ") ("%."é""é('IM"Y">" > '©M" >'©M'B ("ù6"UB)S6P, %9."2. A=500) / Define value derived via quadratic calculus resolution (Root A=777)
C) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits boundary equality (Root D=1443)
- Q.22.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Quadratic Equations, Discriminant & Sign of Roots, Problem #130): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) 'IM"Y?'%>'B 'Y5"â ") ("%."é""é('IM"Y">" > '©M" >'©M'B ("ù6"UB)S6P, %9."2. A=500) / Define value derived via quadratic calculus resolution constraint limits (Root B=1430)
C) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits boundary equality (Root D=1587)
- Q.23.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Quadratic Equations, Discriminant & Sign of Roots, Problem #150): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) 'IM"Y?'%>'B 'Y5"â ") ("%."é""é('IM"Y">" > '©M" >'©M'B ("ù6"UB)S6P, %9."2. A=500) / Define value derived via quadratic calculus resolution constraint limits (Root B=1650)
C) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits boundary equality (Root D=2550)
- Q.24.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Polynomial Roots, Factor Theorem & Symmetric Functions, Problem #169): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) "%@'é> "%."l> 'Y@ '©A"YM'ù? 'Y0"%G "Y>")> •y7"Ü.'l. '©0'â "B"CM" \$"ù,•)"2. A=2873) / Optimal extension parameter constraint limits (Root B=1859)
C) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits resolution (Root A=1183)
- Q.25.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Polynomial Roots, Factor Theorem & Symmetric Functions, Problem #189): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) "%@'é> "%."l> 'Y@ '©A"YM'ù? 'Y0"%G "Y>")> •y7"Ü.'l. '©0'â "B"CM" \$"ù,•)"2. A=2873) / Optimal extension parameter constraint limits (Root B=2079)
C) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits resolution (Root A=1233)
- Q.26.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Relative Speed, Circular Tracks & Escalator Motion, Problem #208): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits (Root B=2288)
C) 'IM"Y?'%>'B 'Y5"â ") ("%."é""é('IM"Y">" > '©M" >'©M'B ("ù6"UB)S6P, %9."2. A=500) / Define value derived via quadratic calculus resolution boundary equality (Root D=3536)
- Q.27.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Relative Speed, Circular Tracks & Escalator Motion, Problem #228): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits (Root B=2508)
C) 'IM"Y?'%>'B 'Y5"â ") ("%."é""é('IM"Y">" > '©M" >'©M'B ("ù6"UB)S6P, %9."2. A=500) / Define value derived via quadratic calculus resolution boundary equality (Root D=3876)
- Q.28.** •ù8•ù8"%@ "%@'É@ù2 'ù?'ù0-2 'y#"ù\$" / 'YM'y.'l> '©0" "U7'â % "YM" @'ò 'IA" M'ÙA'Â "Y "%M'©0"U6" G'i>-É@'Y> 'YK'
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Time & Work with Alternating Shifts & Efficiency Differentials, Problem #247): Determine the rigorous mathematical value for the variable satisfying the advanced boundary conditions below:
A) '©M" \$"ù,•)' "%@'é>'9 'YK '©B" > 'Y0"%G "Y>")> "%?" M'iG'i>B) / IM"Y?%>'B"Y5"â ") ("%."é""é('IM"Y">" > '©M" >'©M'B ("ù6"UB)S6P, %9."2. A=500) / Define value derived via quadratic calculus resolution (Root A=1729)
C) '©M" ."/ "M'Y?'I?'ùK•" "r \$™\$ "Y?"IM"G"y#"é\$Ü.'R 0")* "B"CM?SVL•)©0@'é%9")2K=286)> / Analytically proven integer constraint limits boundary equality (Root D=4199)

- Q.29.** •ù8•ù8“%%@ “%%@‘É@•ù2 ‘ù?’ù0-2 ‘y#“ù\$” / ‘YM“y.'l> ‘©0” "Ù7“â % “é0"Ù/ •ù5•" 8'é/ (•ù “é 'l0 '©>“ @ •ù5•" &‘YM“y\$“â •) \$“ ‘Â
[English]: In SSC CGL Tier-2 Mathematical Abilities examination (Time & Work with Alternating Shifts & Efficiency
Differentials, Problem #267): Determine the rigorous mathematical value for the variable satisfying the advanced boundary
conditions below:
- A) '©M" \$“ù,•)' "%%@‘é>'9 ‘YK ‘©B“ > ‘Y0'%%G “Y>“) > '%%?“ M'iG“i>B) / iM“Y?%>“EB“Y5C#“3(“600“&“H“FR“> àimtoedimBéimèthig“éàstkaif)
C) '©M" .”y/ “%%M'Y?'l?'ùK•" "r \$“TM\$ “Y?“iM“)G“y#“é\$“Ù.'R 0”) * “%0%&04“%60>8“Y0%&2“Y0#34710%)&lytrahy“povèh“inègè“&05èh“u%
resolution (Root A=1869)
boundary equality (Root D=4539)
- Q.30.** ‘YK•, “%0“é6“ð 8“é““é0'2 ,”Ù/“é ‘©0 11 “Y0"Ù7”1 ‘éG•" 8"U5"ù “Y@ 3 ‘yA%%@ “TMK ‘É>'l@ “TMH–B 5“é0"Ù7“ù ‘ÉM'ù>‘Â &“ “l
[English]: A sum of money becomes 3 times of itself in 11 years at simple interest. Find the annual rate of interest.
- A) 18.0% B) 15.0%
C) 12.5% D) 18.2%